

Expected Return

The probability-weighted average of all possible returns.

$$E(R) = \sum [\text{Probability} \times \text{Return}]$$
$$= P_1R_1 + P_2R_2 + P_3R_3 + \dots + P_nR_n$$

Example: TechCorp Stock

Economic Scenario	Probability (P)	Return (R)	P × R
Boom	0.25	40%	10.00%
Normal Growth	0.50	15%	7.50%
Recession	0.25	-10%	-2.50%
TOTAL	1.00		15.00%

$$E(R) = (0.25 \times 40\%) + (0.50 \times 15\%) + (0.25 \times -10\%)$$
$$= 10\% + 7.5\% + (-2.5\%)$$
$$= 15\%$$

The expected return of TechCorp is 15%.**

Annualized Return (CAGR)

When comparing investments over **different time horizons**, we need to annualize.

$$\text{CAGR} = \left(\frac{[\text{Ending Value}]}{[\text{Beginning Value}]} \right)^{1/n} - 1$$

Where n = number of years

Example:

You invested ₹1,00,000. After **4 years**, it became ₹1,96,000.

$$\text{CAGR} = (1,96,000 / 1,00,000)^{(1/4)} - 1$$

$$= (1.96)^{(0.25)} - 1$$

$$= 1.1832 - 1$$

$$= 0.1832$$

Variance (σ^2) & Standard deviation (σ)

$$\sigma^2 = \sum P_i \times [R_i - E(R)]^2$$

$$\sigma = \sqrt{\sigma^2}$$

Example- In the example of TechCorp, pls calculate Variance of its stock

Solution:

Scenario	P	R	E(R)	R - E(R)	[R - E(R)] ²	P × [R - E(R)] ²
Boom	0.25	40%	15%	25%	625	156.25
Normal	0.50	15%	15%	0%	0	0
Recession	0.25	-10%	15%	-25%	625	156.25
Total						312.50

$$\sigma^2 = 312.50 \text{ (in \% squared terms)}$$

$$\sigma = \sqrt{\sigma^2} = \sqrt{312.50} = 17.68\%$$

Coefficient of Variation (CV)

$$CV = \frac{\text{Standard Deviation } (\sigma)}{\text{Expected Return } E(R)}$$

Measures risk per unit of return — useful for comparing investments with different returns AND risk levels.

Lower CV = Better risk-adjusted investment

Example: Comparing Three Investments

Investment	E(R)	σ	CV
Stock A	20%	25%	1.25
Stock B	12%	10%	0.83 
Stock C	30%	45%	1.50

Covariance (S, M):

$$\text{Cov}(S, M) = \frac{\sum \{(R_m - \bar{R}_m) (R_s - \bar{R}_s)\}}{n-1}$$

$$\text{Variance (market)} = \frac{\sum (R_m - \bar{R}_m)^2}{n-1}$$

Beta (market Risk)

$$\beta = \frac{\text{Covariance}(\text{Stock, Market})}{\text{Variance}(\text{Market})}$$

$$\beta = \frac{\sigma(\text{stock, market})}{\sigma^2(\text{market})}$$

Portfolio Return

$$E(R_p) = w_1E(R_1) + w_2E(R_2) + \dots + w_nE(R_n)$$

$$E(R_p) = \sum w_i \times E(R_i)$$

Where w_i = weight (proportion) of each asset in portfolio

Portfolio Risk

For two assets:

$$\sigma_p = \sqrt{w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2w_1 w_2 \sigma_1 \sigma_2 \rho_{12}}$$

Where:

- σ_1, σ_2 = Standard deviations of assets 1 and 2
- ρ_{12} = Correlation coefficient between assets 1 and 2
- w_1, w_2 = Weights

Given:

- Stock A: $E(R) = 20\%$, $\sigma = 25\%$
- Stock B: $E(R) = 12\%$, $\sigma = 15\%$
- Weights: $w_A = 60\%$, $w_B = 40\%$
- Correlation: $\rho = 0.3$

Step 1: Portfolio Return

$$\begin{aligned} E(R_p) &= (0.6 \times 20\%) + (0.4 \times 12\%) \\ &= 12\% + 4.8\% \\ &= 16.8\% \end{aligned}$$

Step 2: Portfolio Variance

$$\sigma_p^2 = w_A^2 \sigma_A^2 + w_B^2 \sigma_B^2 + 2w_A w_B \sigma_A \sigma_B \rho$$

$$\begin{aligned}\sigma_p^2 &= (0.6)^2(25)^2 + (0.4)^2(15)^2 + 2(0.6)(0.4)(25)(15)(0.3) \\ &= (0.36)(625) + (0.16)(225) + 2(0.6)(0.4)(25)(15)(0.3) \\ &= 225 + 36 + 54 \\ &= 315\end{aligned}$$

Step 3: Portfolio Standard Deviation

$$\sigma_p = \sqrt{315} = 17.75\%$$

The CAPM Formula

$$E(R_i) = R_f + \beta_i \times [E(R_m) - R_f]$$